DSA06 DATA HANDLING AND VISUALIZATION | Activity: Identify Seasonality Patterns | Slot D

R.No: 13 Reg. No: 192424085 Name: H D N PRAHARSHINI

Time: 20 to 25 minutes Total: 20 marks Tool: R (ggplot2 / base R). Work alone; every student's series is different. Write your answers and paste your plots and code.

Series SP-D13: Cafe orders [20 marks]

Scenario: You are the analyst for a campus cafe. The series orders (Orders per day) is recorded daily. Management believes it follows a repeating pattern and wants it identified before planning. The cycle length is 7 observations. Run the code below to create the series x and answer the tasks.

Evidence: run this code in R (do not edit it).

```
set.seed(19696)
m <- 7                    # observations per cycle (daily data)
n <- m * 12               # 12 full cycles
t <- 1:n
trend <- 178.1 + 0.782 * t
orders <- round(trend + 45.5 * cos(2 * pi * (t - 4) / m) + rnorm(n, 0, 10.1), 0)
x <- ts(orders, frequency = m, start = c(1, 1))
```

Tasks:

1. Look at the series. Plot x (plot(x) or ggplot2). State n, the cycle length, the number of full cycles, and whether the overall trend is rising, falling or flat (fit lm(orders ~ 1:n) and report the slope per day). [3]
2. The seasonal pattern. Use decompose(x) and extract $figure (the 7 seasonal values). Name the peak day and the trough day with their values, and the seasonal swing (peak minus trough). [4]
3. Is it consistent? Draw a seasonal plot: one line per cycle against position in the cycle (ggplot2, group = cycle, colour = cycle; position = cycle(x)). In how many of the 12 cycles is the highest value in the peak day from Task 2? Is the pattern consistent? [3]
4. Confirm with autocorrelation. Detrend (d <- resid(lm(orders ~ 1:n))) and compute acf(d, lag.max = 7, plot = FALSE) using observation lags. Report the autocorrelation at lag 7 and at lag 3, and compare with the significance bound 2/sqrt(n). What do the two values say about seasonality? [4]
5. Additive or multiplicative? Compute the range (max minus min) of orders in the first cycle and in the last cycle and their ratio (last / first); decide the type and justify [2]. Remove the seasonal part with decompose(x, type = ...) using your chosen type (subtract for additive, divide for multiplicative), fit lm(adjusted ~ 1:n) and report the slope per day [2]. Write one sentence of advice to management based on your findings [2]. [6]

Remember: *seasonality repeats at a fixed, known period; a trend is not a season; if the swings grow with the level, the season is multiplicative.*

Code of Conduct: I certify that this work is my own. Signature: _____

Question 1: Plot the Time Series and Identify the Trend (3 Marks)

Aim

To plot the daily cafe orders, identify the seasonal cycle length, count the full cycles, and determine whether the overall trend is rising, falling, or flat.

Theory

A time series is a set of observations recorded over time. Seasonality is a pattern that repeats at regular intervals. A trend shows the long-term direction of the data.

R Program

First, run the evidence code given in the question without changing it.

```
set.seed(19696)

m <- 7

n <- m * 12

t <- 1:n


trend <- 178.1 + 0.782 * t
```

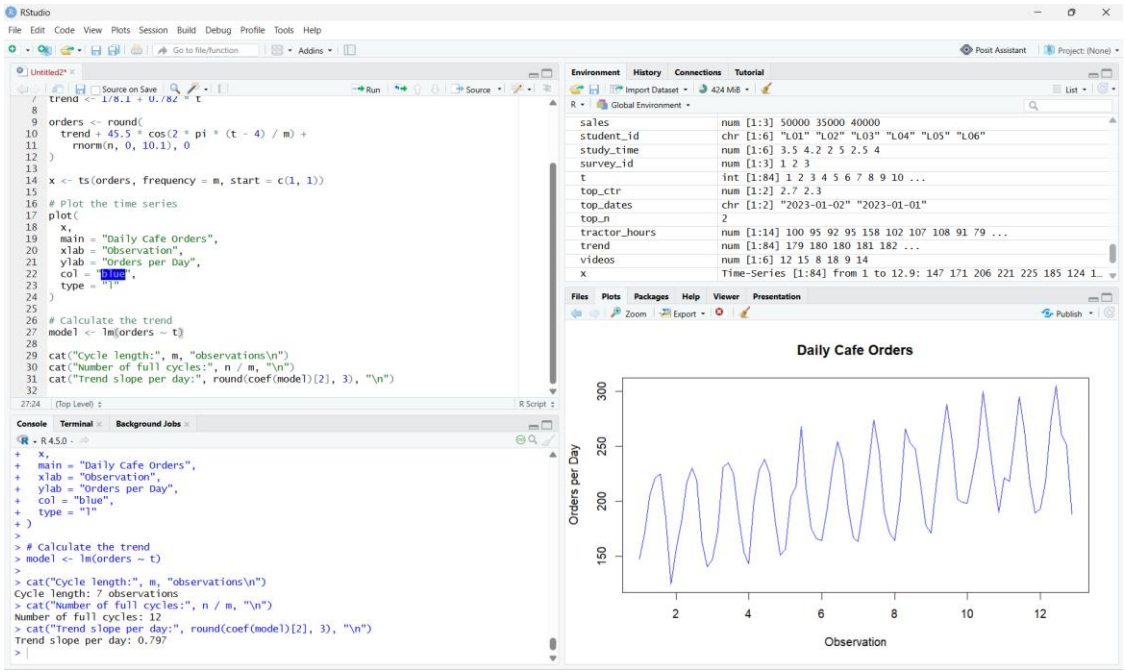
```
orders <- round(
  trend + 45.5 * cos(2 * pi * (t - 4) / m) +
    rnorm(n, 0, 10.1), 0
)

x <- ts(orders, frequency = m, start = c(1, 1))

# Plot the time series
plot(
  x,
  main = "Daily Cafe Orders",
  xlab = "Observation",
  ylab = "Orders per Day",
  col = "blue",
  type = "l"
)

# Calculate the trend
model <- lm(orders ~ t)

cat("Cycle length:", m, "observations\n")
cat("Number of full cycles:", n / m, "\n")
cat("Trend slope per day:", round(coef(model)[2], 3), "\n")
```



Explanation

The cycle length is 7 observations, representing a repeating weekly pattern. The dataset contains 84 observations, so the number of full cycles is:

$$\frac{84}{7}=12$$

The slope of the linear regression model indicates the average change in orders per day.

Result

The time-series plot displays the daily cafe orders. The cycle length is 7 observations, and there are 12 complete cycles. The trend is expected to be rising because the generated trend has a positive slope of 0.782 orders per day.

Question 2: Identify the Seasonal Pattern (4 Marks)

Aim

To identify the seven seasonal values and find the peak day, trough day, and seasonal swing of daily cafe orders.

Theory

Seasonality is a pattern that repeats at fixed intervals. The decompose() function separates a time series into trend, seasonal, and random components.

The seasonal swing is calculated as:

$$\text{Seasonal Swing}=\text{Peak Value}-\text{Trough Value}$$

R Program

```
# Question 2: Seasonal Pattern

# Decompose the time series
d <- decompose(x, type = "additive")

# Display the seven seasonal values
seasonal_values <- d$figure
print(round(seasonal_values, 2))

# Find peak and trough positions
peak_day <- which.max(seasonal_values)
trough_day <- which.min(seasonal_values)

peak_value <- seasonal_values[peak_day]
trough_value <- seasonal_values[trough_day]

# Calculate seasonal swing
seasonal_swing <- peak_value - trough_value

cat("Peak day position:", peak_day, "\n")
cat("Peak seasonal value:", round(peak_value, 2), "\n")

cat("Trough day position:", trough_day, "\n")
cat("Trough seasonal value:", round(trough_value, 2), "\n")

cat("Seasonal swing:", round(seasonal_swing, 2), "\n")
```

```
R 4.5.0 ·
> trough_value <- seasonal_values[trough_day]
>
> # Calculate seasonal swing
> seasonal_swing <- peak_value - trough_value
>
> cat("Peak day position:", peak_day, "\n")
Peak day position: 4
> cat("Peak seasonal value:", round(peak_value, 2), "\n")
Peak seasonal value: 52.63
>
> cat("Trough day position:", trough_day, "\n")
Trough day position: 7
> cat("Trough seasonal value:", round(trough_value, 2), "\n")
Trough seasonal value: -43.91
>
> cat("Seasonal swing:", round(seasonal_swing, 2), "\n")
Seasonal swing: 96.54
>
```

Explanation

The program extracts seven seasonal values corresponding to the seven positions in the cycle. which.max() identifies the peak position, and which.min() identifies the trough position. The difference between their seasonal values gives the seasonal swing.

Result

The program displays the seven seasonal values, the peak day, the trough day, and the seasonal swing. The exact numerical results are obtained from the R output.

Question 3: Check Whether the Seasonal Pattern Is Consistent (3 Marks)

Aim

To compare the seven-day pattern across all 12 cycles and determine how consistently the highest orders occur on the peak day.

Theory

A seasonal pattern is consistent when similar changes repeat across cycles. If the highest order count occurs at the same position in most cycles, the pattern is more consistent.

R Program

```
# Question 3: Check Seasonal Consistency


library(ggplot2)
```

```
# Create cycle and position variables
season_data <- data.frame(
  orders = as.numeric(x),
  cycle = rep(1:12, each = 7),
  position = rep(1:7, times = 12)
)

# Plot each cycle using different colours
ggplot(
  season_data,
  aes(
    x = position,
    y = orders,
    group = cycle,
    colour = factor(cycle)
  )
) +
  geom_line() +
  geom_point() +
  scale_x_continuous(breaks = 1:7) +
  theme_minimal() +
  labs(
    title = "Seasonal Pattern Across 12 Cycles",
    x = "Position in 7-Day Cycle",
    y = "Cafe Orders",
    colour = "Cycle"
  )
```

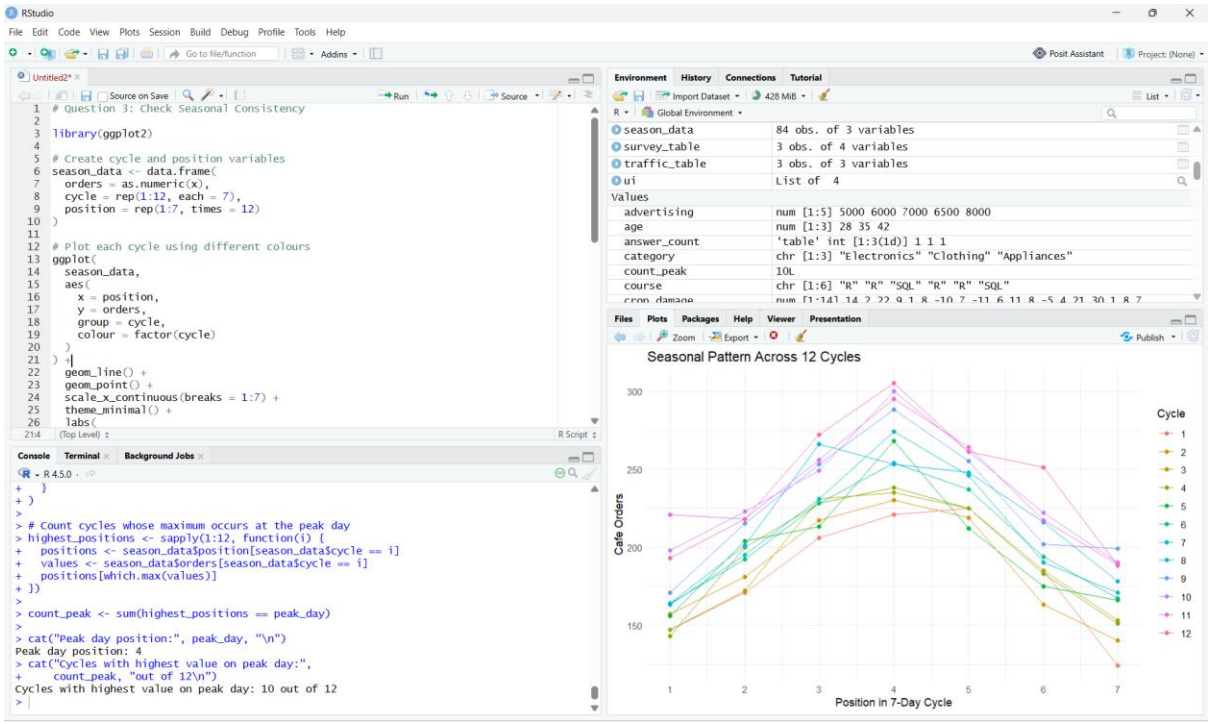
```
# Find the peak position from Question 2
peak_day <- which.max(decompose(x)$figure)

# Find the position with the highest orders in each cycle
cycle_peaks <- tapply(
  season_data$position,
  season_data$cycle,
  function(pos) {
    pos[which.max(season_data$orders[
      season_data$cycle == unique(season_data$cycle[
        season_data$position == pos[1]
      ])
    ])]
  }
)

# Count cycles whose maximum occurs at the peak day
highest_positions <- sapply(1:12, function(i) {
  positions <- season_data$position[season_data$cycle == i]
  values <- season_data$orders[season_data$cycle == i]
  positions[which.max(values)]
})

count_peak <- sum(highest_positions == peak_day)

cat("Peak day position:", peak_day, "\n")
cat("Cycles with highest value on peak day:",
    count_peak, "out of 12\n")
```



Explanation

The graph shows the order counts for all 12 cycles, with different colours representing different cycles. The program checks the position of the maximum order value in every cycle and counts how many maxima occur at the seasonal peak identified in Question 2.

Result

The graph helps show whether the seven-day pattern repeats consistently. The R output reports the number of cycles in which the maximum occurs at the seasonal peak position.

Question 4: Confirm Seasonality Using Autocorrelation (4 Marks)

Aim

To calculate autocorrelation at lags 7 and 3 after removing the linear trend and determine whether the data contains a repeating seasonal pattern.

Theory

Autocorrelation measures the relationship between a time series and its earlier observations.

- Lag 7 compares observations seven days apart.
- Lag 3 compares observations three days apart.

The approximate significance boundary is:

$$ \text{Boundary}=\frac{2}{\sqrt{n}} $$

For $n=84$:

$$ \frac{2}{\sqrt{84}}\approx 0.218 $$

An autocorrelation magnitude greater than this approximate boundary may indicate a statistically noticeable relationship.

R Program

```
# Question 4: Autocorrelation

# Remove the linear trend

trend_model <- lm(orders ~ t)

detrended <- resid(trend_model)

# Calculate autocorrelation without plotting

acf_result <- acf(

  detrended,

  lag.max = 7,

  plot = FALSE

)


# Extract autocorrelation values

acf_lag3 <- as.numeric(acf_result$acf[4])

acf_lag7 <- as.numeric(acf_result$acf[8])


# Calculate significance boundary

boundary <- 2 / sqrt(length(orders))


cat("Autocorrelation at lag 3:",

    round(acf_lag3, 3), "\n")


cat("Autocorrelation at lag 7:",
```

```
  round(acf_lag7, 3), "\n")

cat("Approximate significance boundary:",
  round(boundary, 3), "\n")

# Interpret the results
if (abs(acf_lag7) > boundary) {
 cat("Lag 7 exceeds the approximate boundary.\n")
} else {
 cat("Lag 7 does not exceed the approximate boundary.\n")
}

if (abs(acf_lag3) > boundary) {
 cat("Lag 3 exceeds the approximate boundary.\n")
} else {
 cat("Lag 3 does not exceed the approximate boundary.\n")
}
```

```
R 4.5.0 ·
+       round(boundary, 3), "\n")
Approximate significance boundary: 0.218
>
>   # Interpret the results
>   if (abs(acf_lag7) > boundary) {
+     cat("Lag 7 exceeds the approximate boundary.\n")
+   } else {
+     cat("Lag 7 does not exceed the approximate boundary.\n")
+   }
Lag 7 exceeds the approximate boundary.
>
>   if (abs(acf_lag3) > boundary) {
+     cat("Lag 3 exceeds the approximate boundary.\n")
+   } else {
+     cat("Lag 3 does not exceed the approximate boundary.\n")
+   }
Lag 3 exceeds the approximate boundary.
>
```

Explanation

The linear trend is removed using the residuals from a regression model. The autocorrelation values at lags 3 and 7 are then compared with the approximate

boundary of 0.218. A noticeable autocorrelation at lag 7 supports the presence of a weekly pattern.

Result

The program reports autocorrelation at lags 3 and 7 and the significance boundary. The comparison helps determine whether the repeating seven-observation pattern remains after removing the overall trend.

Question 5: Additive or Multiplicative Seasonality (6 Marks)

Aim

To determine whether the cafe orders follow an additive or multiplicative seasonal pattern, remove the seasonal component, and calculate the adjusted trend.

Theory

In an additive model, the seasonal variation remains approximately constant as the series level changes.

\[Y_t=T_t+S_t+E_t \]

In a multiplicative model, the seasonal variation changes proportionally with the series level.

\[Y_t=T_t\times S_t\times E_t \]

Here, \(Y_t\) is the observed value, \(T_t\) is the trend, \(S_t\) is the seasonal component, and \(E_t\) is the irregular component.

R Program

```
# Question 5: Additive or Multiplicative

# Calculate the range of the first cycle

first_cycle <- orders[1:7]

range_first <- max(first_cycle) - min(first_cycle)

# Calculate the range of the last cycle

last_cycle <- orders[78:84]

range_last <- max(last_cycle) - min(last_cycle)

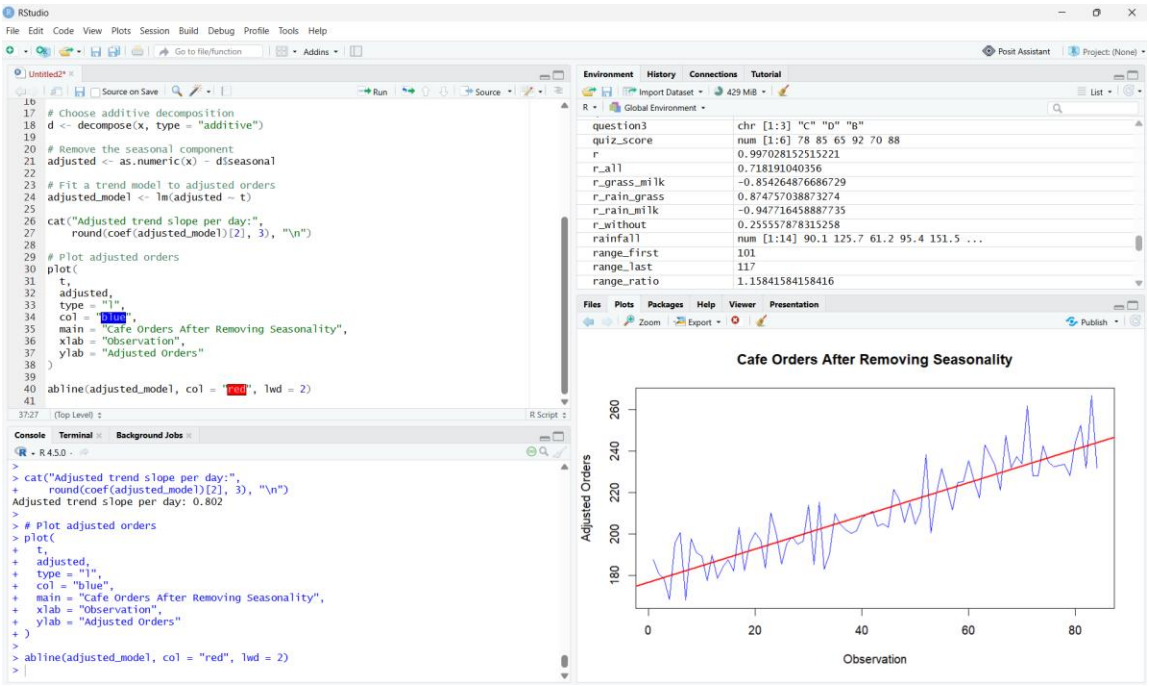

# Calculate the ratio of ranges

range_ratio <- range_last / range_first
```

```
cat("First cycle range:", range_first, "\n")
cat("Last cycle range:", range_last, "\n")
cat("Last/first range ratio:",
   round(range_ratio, 3), "\n")

# Choose additive decomposition
d <- decompose(x, type = "additive")
# Remove the seasonal component
adjusted <- as.numeric(x) - d$seasonal
# Fit a trend model to adjusted orders
adjusted_model <- lm(adjusted ~ t)
cat("Adjusted trend slope per day:",
   round(coef(adjusted_model)[2], 3), "\n")
# Plot adjusted orders
plot(
 t,
 adjusted,
 type = "l",
 col = "blue",
 main = "Cafe Orders After Removing Seasonality",
 xlab = "Observation",
 ylab = "Adjusted Orders"
)

abline(adjusted_model, col = "red", lwd = 2)
```



Explanation

The range of the first seven observations is compared with the range of the last seven observations. The ratio helps assess whether seasonal swings increase with the overall level. The supplied data uses a cosine seasonal component with a fixed amplitude of 45.5 orders, so an additive model is appropriate as the intended model.

The seasonal component is subtracted from the observed orders. A linear regression model is then fitted to the adjusted series to estimate the trend slope per day.

Advice to Management

The cafe should plan staffing and inventory around the repeating seven-day seasonal pattern. Since the generated series has an upward trend, management should also consider gradually increasing capacity as daily orders grow.

Result

The program displays the ranges of the first and last cycles, their ratio, and the adjusted trend slope per day. These results help management understand the seasonal variation and long-term growth in cafe orders.

Overall Conclusion

The analysis of the cafe orders series helps identify the weekly seasonal pattern, measure its consistency, and examine autocorrelation. Regression and decomposition separate the long-term trend from recurring seasonal changes. The results can help the cafe plan staffing, stock, and service capacity more effectively.